

Akshay Mohan Gupta

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Education

University of Waterloo

Sep 2026 - April 2031

Bachelor of Software Engineering Co-op

Waterloo, Canada

- GPA: 4.0/4.0, **\$22,000** Scholarship (National Mathematics, David Gerard Conroy, President's Distinction)
- Relevant Courses: Data Structures & Algorithms, OOP, Software Design, Computer Architecture, Linear Algebra.

Experience

Machine Learning Engineer

Jun 2026 - Sep 2026

Royal Bank of Canada (RBC)

Toronto, ON

- Built a prompt difficulty classifier that **scores incoming queries** for complexity before routing to various LLMs.
- Used Qwen 3.5 0.8B **fine tuned via Llama Factory** on an internal RBC dataset to optimize routing capabilities.
- Incorporated benchmarks/parameters including SWE-bench, TruthfulQA, Finance Agent, cost, latency, and accuracy.
- Used GPT-5 mini as a **LLM judge**, to evaluate the results which showcased a **92% accuracy**.
- Incorporated as a headless application intercepting **10,000+ requests daily**, projected to save **\$100,000/year**.
- Integrated a **Angular frontend** and **Flask backend** within RBC's chatbot so employee's know the optimal LLMs.

Part Time Machine Learning Researcher

Jun 2026 - Sep 2026

Royal Bank of Canada (RBC) Borealis

Toronto, ON

- Extending **TabReason** (RBC Borealis, ICML 2025) with DisCO for imbalanced tabular reasoning.
- Used **DisCO** as an alternative to GRPO for imbalanced classification to RL fine-tune Qwen on tabular reasoning.
- Replaced F1 with **PR-AUC** score evaluation to better capture minority-class detection on skewed credit datasets.
- Fine-tuned Qwen on tabular reasoning data using **Llama Factory**, aligning the model with chain-of-thought rationale

Software Engineer

Jun 2025 - Sep 2025

Royal Bank of Canada (RBC)

Toronto, ON

- Built a **RAG-based classification** service on 5K support tickets using **Nomic embeddings + semantic retrieval**.
- **Embedded** natural-language clerk requests and executed **Top-K vector search** to retrieve highest-similarity tickets.
- Passed top-10 candidates and user queries into an LLM reranker to select the ideal support ticket and required forms.
- **Integrated internal RBC APIs** to retrieve account data, then automatically prefill required client forms.
- Used **MongoDB** to store ticket embeddings and linked forms, with **Jenkins/OpenShift** for production deployment.
- Integrated the **React/Flask** workflow directly into **RBC Advisor Assist** for ticket retrieval and form auto-fill.

Award Winning Projects

HRT Macroplace Challenge (103 Globally) 🏆 | Python, TensorFlow, C++, GNN, RL, Nvidia DREAMplace, Docker

- Placed **103 globally** in the Hudson River Trading (HRT) x Partcl Macroplacement challenge.
- **Beat Simulated Annealing by 23.7%** average proxy cost across all 17 IBM benchmarks (peak **+26.5% on ibm02**).
- Built a **GNN + RL pipeline** for VLSI macro placement on 200,000+ node circuits, encoding netlist data as heterogeneous tripartite graphs (macros, nets, ports) with **KNN spatial edges** for congestion-aware reasoning.
- Integrated AutoDMP on **NVIDIA DREAMPlace** as a post-processing legalization engine to resolve macro overlaps.
- **Eliminated O(N²) compute** by vectorizing HPWL on tensor cores and offloading spatial queries to KNN routines

Shopify Minis Hackathon (5th place) 🏆 | Shop Minis SDK, React, TypeScript, Tailwind, Fal AI

- Built **ShopFit** as a Shop Mini that uses AI to recommend outfits based on user data, shopify products, and more.
- Utilized the **Shop Minis SDK** in order to collect user data, access product data, and get relevant products.
- Implemented **AI virtual try-on** (photo + video) and theme-based search layered on Shopify product data.

BramFire Labs (4th @ Amazon x BramHacks) 🏆 | Python, YOLOv11, Typescript, Google Gemini API, Flask, Node.js

- Placed **4th out of 125 teams** for an AI-powered wildfire detection platform via satellite data + drone imagery.
- Built a real-time geospatial data pipeline and fine-tuned a **YOLOv11** model for early fire detection with high recall.
- Developed a full-stack dashboard for live inference, alerting, and heat-map visualization of fire-risk regions.
- Implemented a Twilio notifications system to automatically text/call subscribed users when a fire is detected.

Skills

Languages: Python, C++, Java, Ruby, Go, JavaScript, TypeScript, SQL, Bash

Frameworks/Libraries: PyTorch, TensorFlow, NumPy, Pandas, React, Angular, Flask, Spring Boot, Next.js, Express.js

Tools/Databases: MongoDB, PostgreSQL, Docker, Linux, Jenkins, OpenShift, Llama Factory, LangChain, Google ADK